

ISRC Robotics Event: Drone Navigation Event Rules, Regulations, and Judging Criteria

Rules and Regulations

1. Eligibility

- ☐ **Age Group:** This event is available for participants aged 19-22 only.
- ☐ **Team Size:** Teams must consist of 2-5 members.



2. Event Structure

- ☐ **Rounds:**
 - o **First Round:** All teams will participate in the first round.
 - o **Second Round:** Teams that pass the first round will compete in the second round.
 - o **Final Round:** The top-performing teams from the second round will advance to the final round.
- ☐ **Attempts:** In each round, teams will be given two attempts to complete the course. The best attempt (shortest time with the least penalties) will be considered for scoring.

3. Course Specifications

- ☐ **Course Design:** The course will feature a series of waypoints, obstacles, and tasks, including tunnels, hoops, and landing pads.
- ☐ **Waypoint Distance:** The distance between waypoints will vary from 5 to 20 meters, depending on the course complexity.
- ☐ **Course Length:** The total course length will range from 50 to 100 meters, incorporating both vertical and horizontal challenges.
- ☐ **Obstacles:** The course will include challenges such as narrow passages, moving obstacles, and altitude changes to test the drone's agility and stability.

4. Drone Requirements

- ☐ **Navigation Systems:** Teams are encouraged to use GPS, IMU sensors, cameras, or other navigation aids to follow the course.
- ☐ **Power Supply:** Drones must be powered by batteries. Teams should have spare batteries on hand.

5. Competition Format

- ☐ **Time Trials:** The drone's performance is timed, and the fastest time to complete the course with the fewest penalties is recorded.
- ☐ **Obstacle Navigation:** Drones must navigate through obstacles without deviating from the course. Deviating from the course or colliding with obstacles will incur penalties.
- ☐ **Restarts:** Teams are allowed to restart their drone if it fails to complete a task, but each restart will incur a time penalty.

6. Scoring and Penalties

- ❑ **Time Scoring:** The drone with the shortest time and fewest penalties in each round will receive the highest score.
- ❑ **Penalty Points:**
 - **Missing a Waypoint:** +10 seconds
 - **Colliding with an Obstacle:** +5 seconds
 - **Restarting the Drone:** +15 seconds
- ❑ **Final Scoring:** The best score from the two attempts in each round will be considered for advancing to the next round and for final scoring.

7. Safety Regulations

- ❑ **Drone Safety:** Drones must be designed safely with no sharp edges or hazardous materials.
- ❑ **Course Safety:** The course area must be kept clear of all participants and spectators during the runs.
- ❑ **Emergency Stop:** Each drone must have an accessible emergency stop mechanism.

Judging Criteria for Drone Navigation Event

Scoring Out of 1000 Points:

1. Innovation and Originality (200 points)

- ❑ **Creative Navigation Techniques (80 points):** Evaluate the originality of the drone's design and its approach to navigating the course, including the use of innovative sensors or algorithms.
- ❑ **Advanced Problem-Solving (60 points):** Assess the drone's ability to handle unexpected challenges or deviations in the course.
- ❑ **Novelty in Design (60 points):** Consider the uniqueness of the drone's physical design and its potential to introduce new concepts in aerial robotics.

2. Technical Mastery (250 points)

- ❑ **Precision in Navigation (100 points):** Evaluate the accuracy with which the drone follows the course, maintaining consistent alignment and altitude.
- ❑ **Sensor Integration (80 points):** Assess the effectiveness of the sensors used for navigation, including their calibration and response to course variations.
- ❑ **Robustness and Durability (70 points):** Evaluate the drone's ability to function reliably under various conditions, including the presence of obstacles or changes in altitude.

3. Practical Applicability and Global Impact (150 points)

- ❑ **Real-World Applications (70 points):** Consider the potential of the drone's design to be applied in real-world scenarios, such as search and rescue missions or industrial inspections.

- **Environmental Considerations (50 points):** Evaluate the sustainability of the drone, including energy efficiency and material usage.
- **Scalability (30 points):** Assess the potential for the drone's design to be scaled for larger or more complex applications.

4. Presentation Excellence (200 points)

- **Technical Explanation (100 points):** Evaluate the team's ability to explain the technical aspects of their drone, including sensor integration, programming, and design choices.
- **Demonstration Quality (70 points):** Assess the effectiveness of the live demonstration, including the drone's performance and the clarity of the team's presentation.
- **Team Communication (30 points):** Consider how well the team collaborates and communicates their ideas during the presentation.

5. Judges' Challenge (100 points)

- **Real-Time Problem Solving (50 points):** Present a real-time challenge to the teams, such as modifying the course or introducing a new obstacle, and evaluate their ability to adapt.
- **Cross-Disciplinary Collaboration (30 points):** Assess how well the team integrates knowledge from different STEM fields, such as engineering, programming, and design.
- **Time-Constrained Innovation (20 points):** Evaluate the team's ability to innovate under time constraints during the challenge.

6. Drone Design and Functionality (100 points)

- **Mechanical Design (40 points):** Assess the structural design, including the materials used, compactness, and overall build quality of the drone.
- **Programming and Control (40 points):** Evaluate the drone's programming logic, control algorithms, and efficiency in executing tasks.
- **Adaptability (20 points):** Consider the drone's ability to adapt to different course conditions, including speed adjustments and obstacle avoidance.